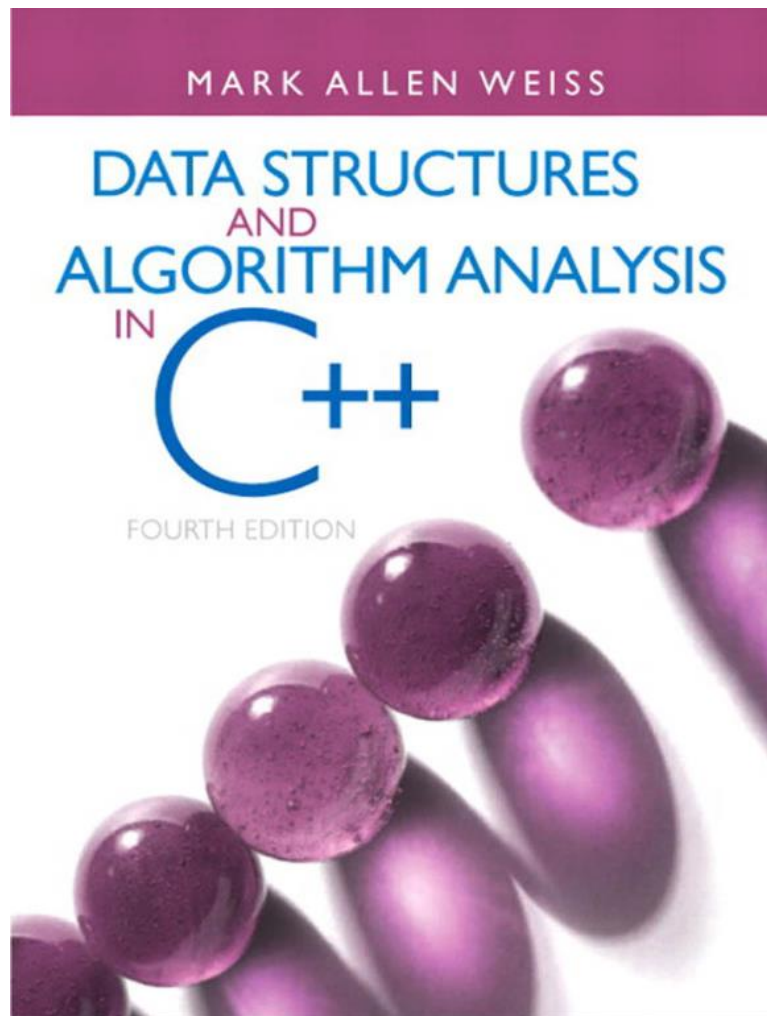


Assignment 3 Inclass : From book “Data Structures and Algorithms Analysis in C++”
by Mark Allen Weiss, p.283



Exercises

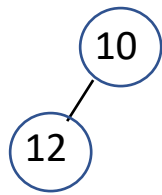
6.1 Can both `insert` and `findMin` be implemented in constant time?

6.2  a. Show the result of inserting 10, 12, 1, 14, 6, 5, 8, 15, 3, 9, 7, 4, 11, 13, and 2, one at a time, into an initially empty binary heap.
b. Show the result of using the linear-time algorithm to build a binary heap using the same input.

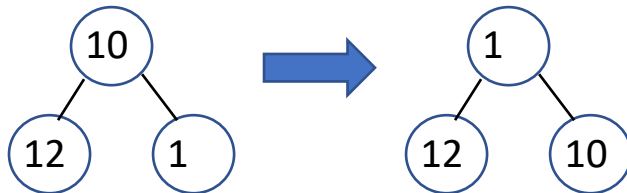
 6.3 Show the result of performing three `deleteMin` operations in the heap of the previous exercise.

Solution:

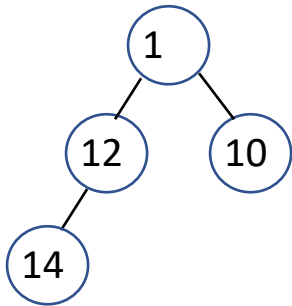
First step: From an empty binary heap, insert 10 then 12 :Check structure prop. and order prop.



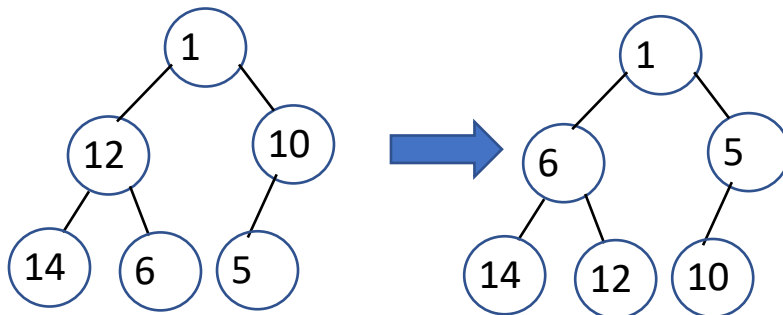
2nd step: Insert 1: Check structure prop. and order prop. Swap 10 1



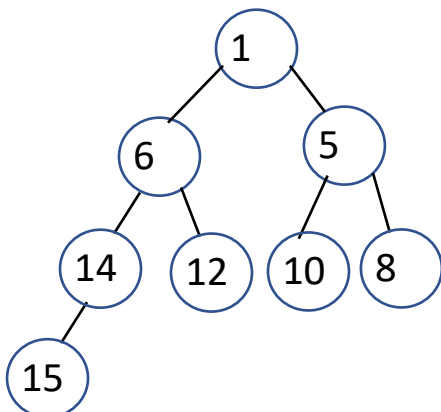
3rd step: Insert 14: Check structure prop. and order prop. No change.



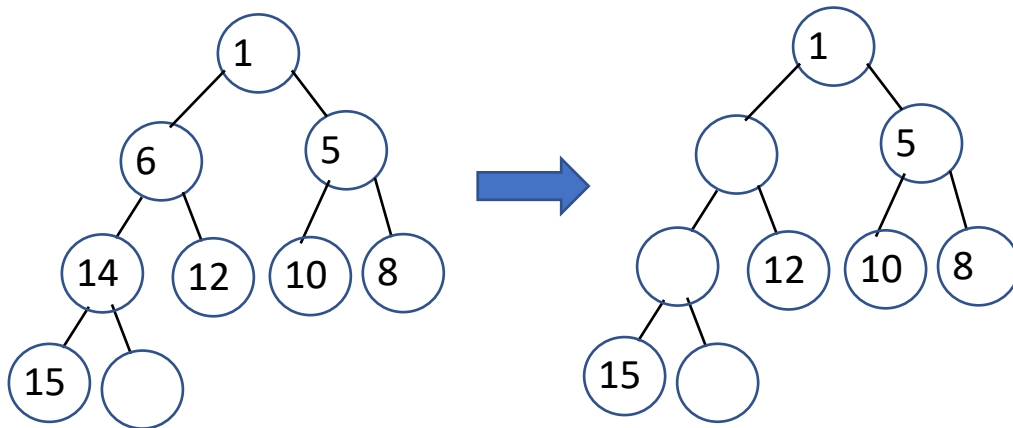
4th step: Insert 6 then 5: Check structure prop. and order prop. Swap 12 6, 10 5.



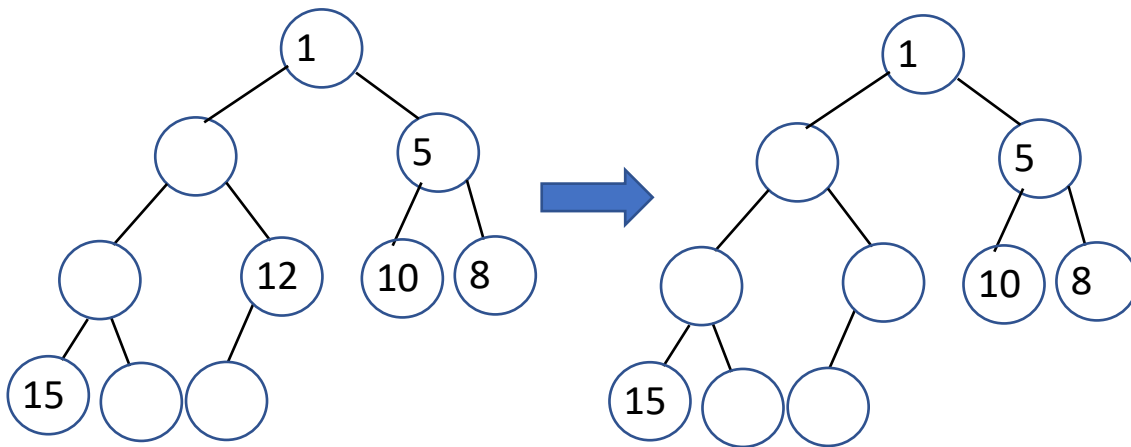
5th step: Insert 8 then 15: Check structure prop. and order prop. No change.



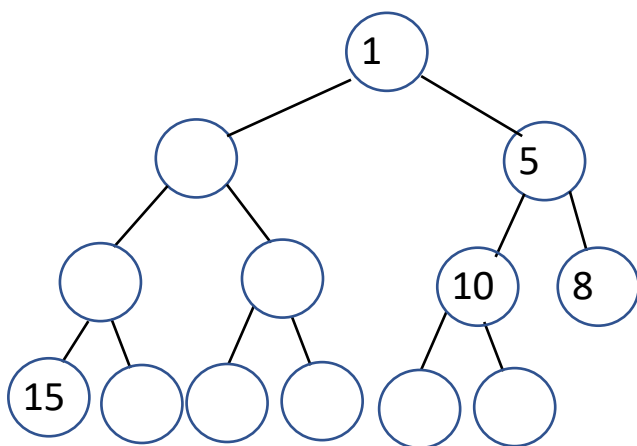
6th step: Insert 3: Check structure prop. and order prop.



7th step: Insert 9: Check structure prop. and order prop.



8th step: Insert 7 then 4 then 11: Check structure prop. and order prop.



9th step: Insert 13 then 2: Check structure prop. and order prop.

Finally: array of binary heap after insert all elements already.

